

#This code will be used to demonstrate theorem 6.17. We will demonstrate that $Q1 = \frac{x^2}{s^2}$ is decreasing along the γ_1 curve; (Part 1 — locating potential critical points)

#Define Q1;

$$Q1 := \frac{x^2 \cdot ((z + w)^2 - y^2)}{y^2 \cdot z \cdot w} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \left. \right\} :$$

q1 := simplify((expand(subs(parametrization, Q1)))) ;

$$q1 := \frac{4 (a b + 1)^2 (-b + a)^2}{(a^2 + 1) (b^2 + 1) (a b - 1)^2} \quad (1)$$

#Let us denote by $D1 = \frac{da}{db}$. By the chain rule, the derivative of Q1 is zero if and only if

$$D1 := - \frac{\text{diff}(q1, b)}{\text{diff}(q1, a)} :$$

#Define r1

$$\begin{aligned} r1 := & \left(a^{32} b^{16} + b^{16} + (-4 b^{18} + 8 b^{16} - 4 b^{14}) a^{30} + (6 b^{20} - 32 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^{28} \right. \\ & + (72 b^{20} - 76 b^{18} + 8 b^{16} + 692 b^{14} + 72 b^{12} - 1024 b^{10}) a^{26} + (b^{32} + 8 b^{30} + 20 b^{28} \\ & - 120 b^{26} - 3426 b^{24} - 7288 b^{22} - 125524 b^{20} - 222104 b^{18} - 279672 b^{16} - 317080 b^{14} \\ & - 157780 b^{12} - 32376 b^{10} + 1950 b^8 + 1672 b^6 + 20 b^4 + 8 b^2 + 1) a^{16} + (-1016 b^{26} \\ & - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\ & + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\ & + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\ & + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\ & - 4 b^{14}) a^2 - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 - 2) (b^2 + 1)^4 a^{23} + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} \\ & - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 \\ & + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \\ & + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} \\ & + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\ & + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} \\ & - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 \\ & + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 \right. \\ & + 2369 b^6 - 699 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 \right. \\ & - 528 b^4 + \frac{4168}{9} b^2 + 1 \left. \right) b^8 a^{24} - 1024 b^{15} (b^2 + 1)^4 a^7 - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 \right. \\ & - 1 \left. \right) b^{11} (b^2 + 1)^4 a^{11} - 1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 \\ & - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} \\ & + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left(b^8 - \frac{52}{7} b^6 \right. \end{aligned}$$

$$+ \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \Big) (b^2 + 1)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) b^7 (b^2 + 1)^4 a^{21} \Big) :$$

:

#Using the implicit function theorem, compute $\frac{da}{db} = D2$ in terms of a and b

$$D2 := - \frac{\text{diff}(r1, b)}{\text{diff}(r1, a)} :$$

#We construct the curve $D2 - D1$ where its zeros describe points where the derivative of $\frac{x^2}{s^2}$ is zero

$$f := D2 - D1 :$$

#Calculate the numerator of f

$$\tilde{f} := \text{numer}(f) :$$

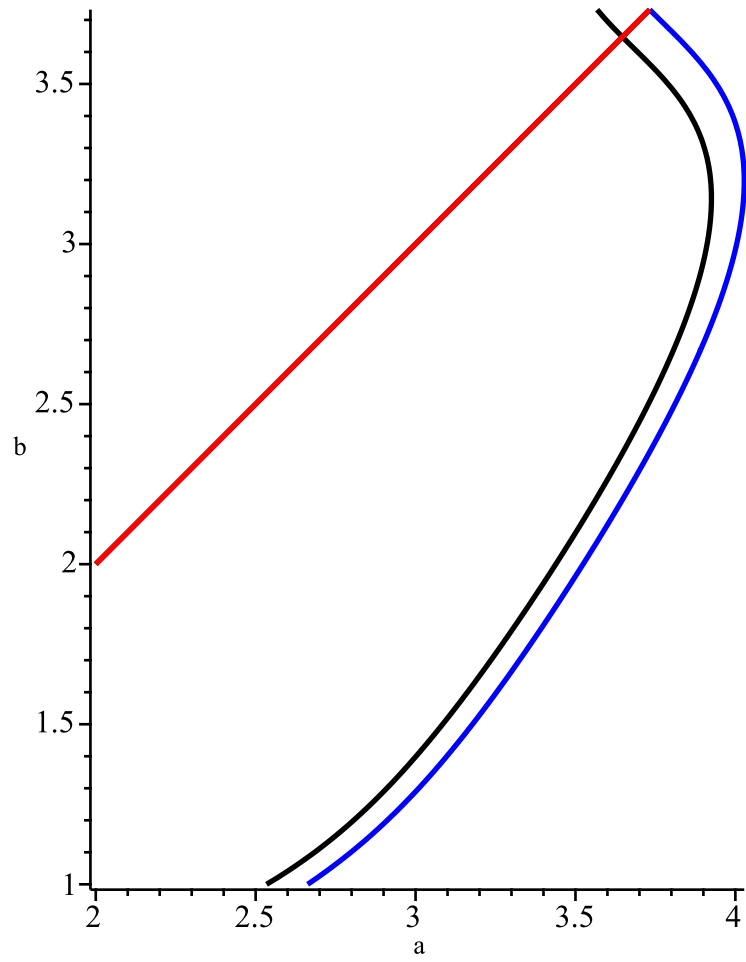
#Plot the curves $r1=0$ and $\tilde{f}=0$ in region $A1$.

$$r := a - b :$$

with(plots) :

```
implicitplot([ r1=0,  $\tilde{f}=0$ ,  $r=0$  ], a=2 .. 5, b=1 .. (2 + sqrt(3)),
  scaling=constrained,
  gridrefine=2,
  thickness=2,
  color=[blue, black, red],
  labels=["a", "b"],
  title="Graphic for  $r1=0$  and  $\tilde{f}=0$  in  $A1$ ");
```

Graphic for $r_1=0$ and $\tilde{f}=0$ in A1



#Let's build the DescartesSingChanges function that counts the number of sign changes between the coefficients of a given polynomial.

```
DescartesSignChanges := proc( polynomial, variable)  
  local coeffs, nonzero_coeffs, s, i, n;  
  
  # Get the list of coefficients from highest degree to constant term  
  coeffs := [seq( coeff( polynomial, variable, i ), i = degree( polynomial, variable ) .. 0, -1 )];  
  
  # Remove zero coefficients (they do not count in Descartes' Rule)  
  nonzero_coeffs := remove( c → c = 0, coeffs );  
  
  # Count the number of sign changes  
  n := nops( nonzero_coeffs );  
  s := 0;  
  for i from 1 to n - 1 do  
    if signum( nonzero_coeffs[ i ] ) * signum( nonzero_coeffs[ i + 1 ] ) = -1 then  
      s := s + 1;  
    end if;  
  end do;  
  
  return s;  
end proc;
```

#Compute the resultant between $r1$ and $\tilde{f}$ with respect to a

$R1 := \text{resultant}(r1, \tilde{f}, a) :$

#Compute the factors of the resultant $R1$ and count how many there are

```
dgb := proc(f)
  if type(f, '^') then
    degree(op(1,f), b);
  else
    degree(f, b);
  end if;
end proc;

bfator := sort(
  [op(factor(R1))],
  (f, g) → dgb(f) < dgb(g)
) :
```

$\text{nops}([op(factor(R1))])$;

14

(2)

#We list below the factors of the resultant $R1$, except for the factor 14 because it is too large. When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[1, 2 + \sqrt{3})$ where γ_1 is defined. For the $rb14$ factor, because of its high degree, we will use Moebius transformations as it requires less processing power.

$rb1 := bfator[1];$

$rb1 := 104886820303173450314022202022447947551802467435937792$ (3)

$rb2 := bfator[2];$

$rb2 := b^{672}$ (4)

$rb3 := bfator[3];$

$rb3 := (b^2 + 4b + 1)^2$ (5)

$rb4 := bfator[4];$

$rb4 := (b^2 - 4b + 1)^2$ (6)

$rb5 := bfator[5];$

$rb5 := (3b^2 - 1)^3$ (7)

$rb6 := bfator[6];$

$rb6 := (b^2 + 1)^{416}$ (8)

$rb7 := bfator[7];$

$\text{sturm}(\text{sturmseq}(rb7, b), b, 1, 2 + \sqrt{3})$; #number of roots in $[1, 2 + \sqrt{3})$

$$rb7 := (b^4 - 4b^3 + 18b^2 - 4b + 1)^2$$

$$0 \quad (9)$$

$rb8 := bfator[8];$

$$rb8 := (b^4 + 4b^3 + 18b^2 + 4b + 1)^2 \quad (10)$$

$rb9 := bfator[9];$

$sturm(sturmseq(rb9, b), b, 1, 2 + \sqrt{3});$ #number of roots in $[1, 2 + \sqrt{3})$

$$rb9 := (3b^4 - 6b^2 + 7)^3$$

$$0 \quad (11)$$

$rb10 := bfator[10];$

$sturm(sturmseq(rb10, b), b, 1, 2 + \sqrt{3});$ #number of roots in $[1, 2 + \sqrt{3})$

$$rb10 := b^6 - 45b^4 + 3b^2 - 15$$

$$0 \quad (12)$$

$rb11 := bfator[11];$

$$rb11 := b^6 + 51b^4 + 3b^2 + 17 \quad (13)$$

$rb12 := bfator[12];$

$sturm(sturmseq(rb12, b), b, 1, 2 + \sqrt{3});$ #number of roots in $[1, 2 + \sqrt{3})$

$$rb12 := (65b^6 - 141b^4 + 195b^2 - 47)^3$$

$$0 \quad (14)$$

$rb13 := bfator[13];$

$sturm(sturmseq(op(1, rb13), b), b, 1, 2 + \sqrt{3});$ #number of roots in $[1, 2 + \sqrt{3})$

$$rb13 := (374017b^{48} - 1082256b^{46} - 2821284b^{44} + 38273024b^{42} - 137706774b^{40} \\ - 25011408b^{38} + 2222925500b^{36} - 11137697376b^{34} + 34666181511b^{32} \\ - 79729549344b^{30} + 140202896376b^{28} - 190367984256b^{26} + 204746625068b^{24} \\ - 177535429920b^{22} + 126486144024b^{20} - 74566977600b^{18} + 36572797215b^{16} \\ - 14813913168b^{14} + 4932365356b^{12} - 1327146624b^{10} + 292102794b^8 - 53430800b^6 \\ + 8765868b^4 - 983136b^2 + 45241)^2$$

$$0 \quad (15)$$

#The factor $rb14$ has degree 376, so we will not show it, just compute how many zeros it has in the desired range.

$rb14 := bfator[14];$

$degree(rb14, b);$

$$376 \quad (16)$$

#From previous tests, using the Moebius transformation on the $rb14$ factor was more efficient when

subdividing the interval $[1, 2 + \sqrt{3})$ into three subintervals: $[1,2)$, $[2,3)$ and $[3, 2 + \sqrt{3})$.

#Apply the Möbius transformation on the interval (1, 2), constructing Rb14_1

$$bmoebius1 := \left\{ b = \frac{1 + 2 \cdot d}{1 + d} \right\} :$$

Rb14_1 := collect(numer(expand(subs(bmoebius1, rb14))), d) :

#From here we will analyze the signal variation of all coefficients Rb14_1

DescartesSignChanges(Rb14_1, d);

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#Apply the Möbius transformation on the interval (2, 3), constructing Rb14_2

$$bmoebius2 := \left\{ b = \frac{2 + 3 \cdot d}{1 + d} \right\} :$$

Rb14_2 := collect(numer(expand(subs(bmoebius2, rb14))), d) :

#From here we will analyze the signal variation of all coefficients Rb14_2

DescartesSignChanges(Rb14_2, d);

0

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#Apply the Möbius transformation on the interval (3, 2 + sqrt(3)), constructing Rb14_3

$$bmoebius3 := \left\{ b = \frac{3 + (2 + \sqrt{3}) \cdot d}{1 + d} \right\} :$$

Rb14_3 := collect(numer(expand(subs(bmoebius3, rb14))), d) :

#From here we will analyze the signal variation of all coefficients Rb14_3

DescartesSignChanges(Rb14_3, d);

1

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#Finding possible candidates for zeros of the factor rb13

fsolve(rb13=0, b, 1 .. (2 + sqrt(3)));

1.748533112, 3.464457702

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#We now repeat the same process for the resultant of r_1 with $\tilde{f}$, in relation to b

$R2 := \text{resultant}(r_1, \tilde{f}, b) :$

#Compute the factors of the resultant $R2$ and count how many there are

```

dg(a) := proc(f)
  if type(f, '^') then
    degree(op(1,f), a);
  else
    degree(f, a);
  end if;
end proc;

afator := sort(
  [op(factor(R2))],
  (f, g) → dg(a)(f) < dg(a)(g)
) :

```

$nops([op(factor(R2))])$;

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#We list below the factors of the resultant $R2$, except for the factor 14 because it is too large. When necessary, we will use Sturm's algorithm to check if the factors cancel each other out within the interval $[2, 5)$ where γ_l is defined. For the ra_{14} factor, because of its high degree, we will use Moebius transformations as it requires less processing power.

$ra_1 := \text{afator}[1]$;

$ra_1 := -104886820303173450314022202022447947551802467435937792$ (22)

$ra_2 := \text{afator}[2]$;

$ra_2 := a^{672}$ (23)

$ra_3 := \text{afator}[3]$;

$ra_3 := (a^2 + 4a + 1)^2$ (24)

$ra_4 := \text{afator}[4]$;

$ra_4 := (a^2 - 4a + 1)^2$ (25)

$ra_5 := \text{afator}[5]$;

$ra_5 := (a^2 - 3)^3$ (26)

$ra_6 := \text{afator}[6]$;

$ra_6 := (a^2 + 1)^{416}$ (27)

$ra_7 := \text{afator}[7]$;

$\text{sturm}(\text{sturmseq}(ra_7, a), a, 2, 5)$; #number of roots in $[2, 5)$

$$ra7 := (a^4 - 4a^3 + 18a^2 - 4a + 1)^2$$

$$0 \quad (28)$$

ra8 := afator[8];

$$ra8 := (a^4 + 4a^3 + 18a^2 + 4a + 1)^2$$

$$(29)$$

ra9 := afator[9];

sturm(sturmseq(ra9, a), a, 2, 5); #number of roots in [2,5)

$$ra9 := (7a^4 - 6a^2 + 3)^3$$

$$0 \quad (30)$$

ra10 := afator[10];

sturm(sturmseq(ra10, a), a, 2, 5); #number of roots in [2,5)

$$ra10 := 15a^6 - 3a^4 + 45a^2 - 1$$

$$0 \quad (31)$$

ra11 := afator[11];

$$ra11 := 17a^6 + 3a^4 + 51a^2 + 1$$

$$(32)$$

ra12 := afator[12];

sturm(sturmseq(ra12, a), a, 2, 5); #number of roots in [2,5)

$$ra12 := (47a^6 - 195a^4 + 141a^2 - 65)^3$$

$$0 \quad (33)$$

ra13 := afator[13];

sturm(sturmseq(op(1, ra13), a), a, 2, 5); #number of roots in [2,5)

$$ra13 := (175559a^{48} - 1769112a^{46} + 6749652a^{44} + 9460584a^{42} - 193871370a^{40}$$

$$+ 750687384a^{38} - 111548076a^{36} - 9218817960a^{34} + 37541503329a^{32}$$

$$- 86577381616a^{30} + 144267501288a^{28} - 188241754608a^{26} + 198904792788a^{24}$$

$$- 173518719696a^{22} + 126058415880a^{20} - 76187388496a^{18} + 37964657145a^{16}$$

$$- 15394822392a^{14} + 5027018436a^{12} - 1306370040a^{10} + 287741334a^8 - 59068744a^6$$

$$+ 10970148a^4 - 1480584a^2 + 96383)^2$$

$$0 \quad (34)$$

#The factor ra13 has degree 376, so we will not show it, just compute how many zeros it has in the desired range.

ra14 := afator[14]:

degree(ra14 , a);

#Apply the Möbius transformation on the interval (2, 5), constructing Ra14

$$amoebius := \left\{ a = \frac{2 + 5 \cdot c}{1 + c} \right\} :$$

$$Ra14 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(amoebius, ra14))), c) :$$

#From here we will analyze the signal variation of all coefficients Ra14

$$\text{DescartesSignChanges}(Ra14, c);$$

$$376$$

$$5$$

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#Finding possible candidates for zeros of the factor ra14

$$fsolve(ra14=0, a, 2 .. 5);$$

$$3.192177604, 3.538902695, 3.555974116, 3.643966007, 3.884436541$$

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#Let's rule out that the root $a=2+\sqrt{3}$ does not form a pair with any of the other candidates b

#Apply the Möbius transformation on the interval $[1, 2+\sqrt{3})$, constructing $\hat{r}1$

$$moebius := \left\{ b = \frac{1 + (2 + \sqrt{3})d}{1 + d} \right\} :$$

$\hat{r}1 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(moebius, \text{eval}(r1, a = 2 + \sqrt{3})))), d) :$

#From here we will analyze the signal variation of all coefficients $\hat{r}1$

DescartesSignChanges($\hat{r}1, d$);

1

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$\text{fsolve}(\text{eval}(r1, a = 2 + \sqrt{3})) = 0, b, 1..2 + \sqrt{3})$

2.350546522, 3.732050803

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